Errata and minor updates

Physical Geodesy: Global Elastic Deformation of the Earth

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1. Page 236, from eq (4.40) to eq (4.43):

$$\Pi > \left[1 - \frac{\alpha}{(N+1)N}\right] \cdots \left[1 - \frac{\alpha}{(i-3)(i-4)}\right] \left[1 - \frac{\alpha}{(i-1)(i-2)}\right] \times \left[1 - \frac{\alpha}{(i+1)i}\right] \left[1 - \frac{\alpha}{(i+3)(i+2)}\right] \cdots \left[1 - \frac{\alpha}{(i+l+1)(i+l)}\right] \cdots . \tag{4.40}$$

Take the logarithm on both sides,

$$\begin{aligned} \ln \Pi > {} & \ln \left[1 - \frac{\alpha}{(N+1)N}\right] + \cdots + \ln \left[1 - \frac{\alpha}{(i-3)(i-4)}\right] \\ & + \ln \left[1 - \frac{\alpha}{(i-1)(i-2)}\right] + \ln \left[1 - \frac{\alpha}{(i+1)i}\right] \\ & + \ln \left[1 - \frac{\alpha}{(i+3)(i+2)}\right] + \cdots + \ln \left[1 - \frac{\alpha}{(i+l+1)(i+l)}\right] + \cdots . \end{aligned} \tag{4.41}$$

As $\lim_{l\to\infty}\{\alpha/[(i+l+1)(i+l)]\} = 0$, we have

$$\lim_{i\to\infty} \left\{ \ln \left[1 - \frac{\alpha}{(i+l+1)(i+l)}\right] \Big/ \left[-\frac{\alpha}{(i+l+1)(i+l)}\right] \right\} = 1 , \tag{4.42}$$

which can be considered as an alternative form of the formula $\lim_{x\to 0}[\ln(1+x)]/x = 1$. Hence, the infinite series at the right-hand side of (4.41) converges or diverges in the same way as the series

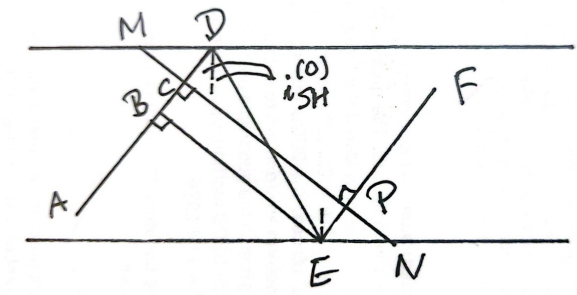
$$\begin{aligned} & \frac{\alpha}{(N+1)N} + \cdots + \frac{\alpha}{(i-3)(i-4)} + \frac{\alpha}{(i-1)(i-2)} + \frac{\alpha}{(i+1)i} \\ & \quad + \frac{\alpha}{(i+3)(i+2)} + \cdots + \frac{\alpha}{(i+l+1)(i+l)} + \cdots , \end{aligned} \tag{4.43}$$

2. Page 280, left hand side of the second line of eqn (4.296):
   Replace "a_k" by "b_k"

3. Page 281, in eqn (4.302):
   Replace "2π" by "4π"

4. Page 283, line 1 below (4.314):
   Change to "This formula holds for arbitrarily large N. Therefore, the positive term series" and change "$\sum_{n=0}^{N}$" to "$\sum_{n=0}^{\infty}$" in (4.315)

5. Page 514, replace Fig. 6.17 by (the original may probably have a copyright issue)



6. Page 753, eqn (9.59) should be

$$\delta s(\mathrm{GR}) = \frac{R\bar{\rho}}{3} \sum_{n=2}^{n_{\max}^{\mathrm{G2}}} \sum_{k=-n}^{n} \frac{2n+1}{1+k'_n} \left[\delta\Phi_n^k(\mathrm{GR}) - \delta\Phi_n^k(\mathrm{GA})\right] \bar{\mathcal{H}}_n^k(\theta,\lambda)\,,$$